

Understanding Continuous Integration and Continuous Delivery/Deployment

This article discusses continuous integration (CI) and continuous delivery/deployment (CD), which are part and parcel of the DevOps software development culture. The goal of all developers is to produce software that is reliable, reusable, extendable, flexible, correct and efficient. DevOps ensures this, with CI and CD as integral parts of the process.

In simple terms, integrate when you commit. Continuous integration implementation doesn't mean less bugs. Rather, it highlights the issues or bugs in the early stages and hence is useful because the earlier in the development cycle you fail, the faster you recover!

The benefits of detecting bugs early

In the case of discovering failures or bugs, it becomes the priority of the respective stakeholders to focus on solving build or continuous integration issues at the earliest and fix the broken build.

Continuous integration (CI) is a popular DevOps practice that requires the development team to commit code into a shared repository (Centralised Version Control or Distributed Version Control) as and when a feature is completed or a bug is fixed. Each commit goes through a build validation process using an automated build process with any automation tool feasible, based on the knowledge or the culture of the organisation.

It is important for the development team to commit frequently whenever a feature is implemented or a bug is fixed. There are still some developers who commit code even if it is not properly tested or is not working fine. There are also instances when code is committed

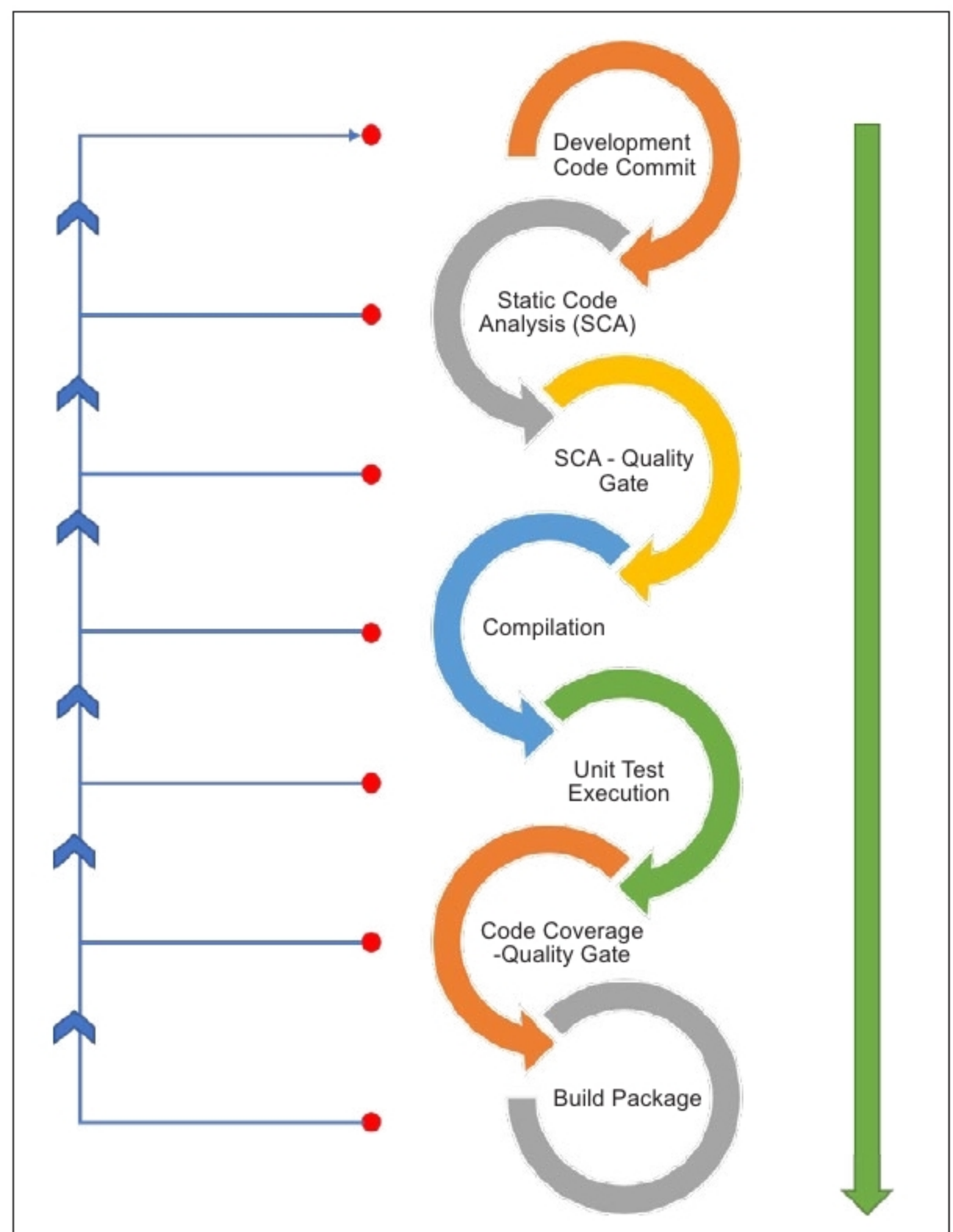


Figure 1: Continuous integration

